

ADITYA VERMA

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks(%)
2021 - 2025	B.TECH in Computer Science and Engineering	Graphic Era University, Dehradun	8.32

EXPERIENCE

Software Engineer, TCS - *Mumbai, MH*

November, 2025 - Current

- Designed and developed scalable backend systems using Python and PySpark as part of the Reserve Bank of India's Centralised Information Management System modernization, enabling the transition from report-based to element-based data architecture for 200+ regulatory workflows across ~15,000 financial institutions.
- Architected and implemented bi-directional data processing frameworks, including ingestion of structured SDMX data and dynamic generation of regulatory reports, powering both real-time query platforms and end-user reporting systems.
- Built high-performance data processing modules leveraging distributed computing (Apache Spark), optimizing data transformation, validation, and mapping logic to ensure system reliability, consistency, and regulatory compliance at scale.
- Engineered robust data pipelines and backend services integrating SQL and Hive, focusing on performance optimization, fault tolerance, and maintainability in large-scale production environments.
- Led the development and optimization of large-scale data migration systems, identifying and resolving system-level defects, ensuring accurate and consistent transformation of legacy datasets into the new architecture.

PROJECTS

WordSphere: A comprehensive blogging application (Aug, 2024 - Sept, 2024) : [Link](#)

- Built a seamless post creation, editing, and deletion functionalities for an intuitive user experience.
- Added advanced filters that improve user accessibility and content discovery within the application. Moreover, elevated the user's creative experience with a powerful TinyMCE editor for crafting compelling blog content.

Self-Driving Car Agent (Sept, 2023 - Dec, 2023) : [Link](#)

- Designed a self-driving car agent using Unity3D's ML-Agents module, focusing on PPO (Proximal Policy Optimization) and SAC (Soft Actor-Critic) models with a decision-making accuracy of 92%.
- Attained an average adaptation time of 3.5 seconds to environmental changes.

Real-Time Vehicle Tracking System (Aug, 2023 - Sept, 2023) : [Link](#)

- Engineered a real-time vehicle tracking application leveraging ReactJS and Tailwind CSS and achieved a 98% real-time update accuracy.
- Innovated a comprehensive and user-friendly interface for vehicle location and status monitoring.

ACHIEVEMENTS

- TCS AI Fridays'** - Secured 1st place amongst 15 teams battling for 7 hours creating a well-documented and working RAG based application for given problem statement in an internal AI Hackathon.
- Graphic Era Coding Competition(ICCC)** - Secured 3rd rank in a coding contest hosted by Graphic Era.
- TechWizard** - Qualified for 2nd round twice for an Inter-University coding contest hosted by Graphic Era University.
- INFathon** - Qualified for Round 2 of a Code-a-thon organized by Informatica.

SKILLS

Technical Skills	Python, Java, JavaScript, HTML/CSS, C++, AI/ML(GenAI, NLP, LLM, RAG)
Database	SQL, MongoDB
Tools & Frameworks	PySpark, ReactJS, ExpressJS, TailwindCSS, Mongoose, Git, Docker
Cloud Technologies	Google Cloud Platform

CERTIFICATIONS

- Certified Salesforce Agentblazer Innovator
- Certified AWS GenAI Practitioner
- Google Cloud - Introduction to GenAI
- Google Cloud Computing Foundations: Cloud Computing Fundamentals